

1D CNN — Architecture Variant Comparison

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Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic
Total trials	360 (train pool 324, hold-out 36)
Classes	6: Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap
Sensor channels	144 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	per_trial_zscore
CV folds · shuffle	7 · True
Augmentation	on · copies=6 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout
Epochs · batch size	60 · 16

Sensor grid (✓ enabled, ✗ disabled, — not present)

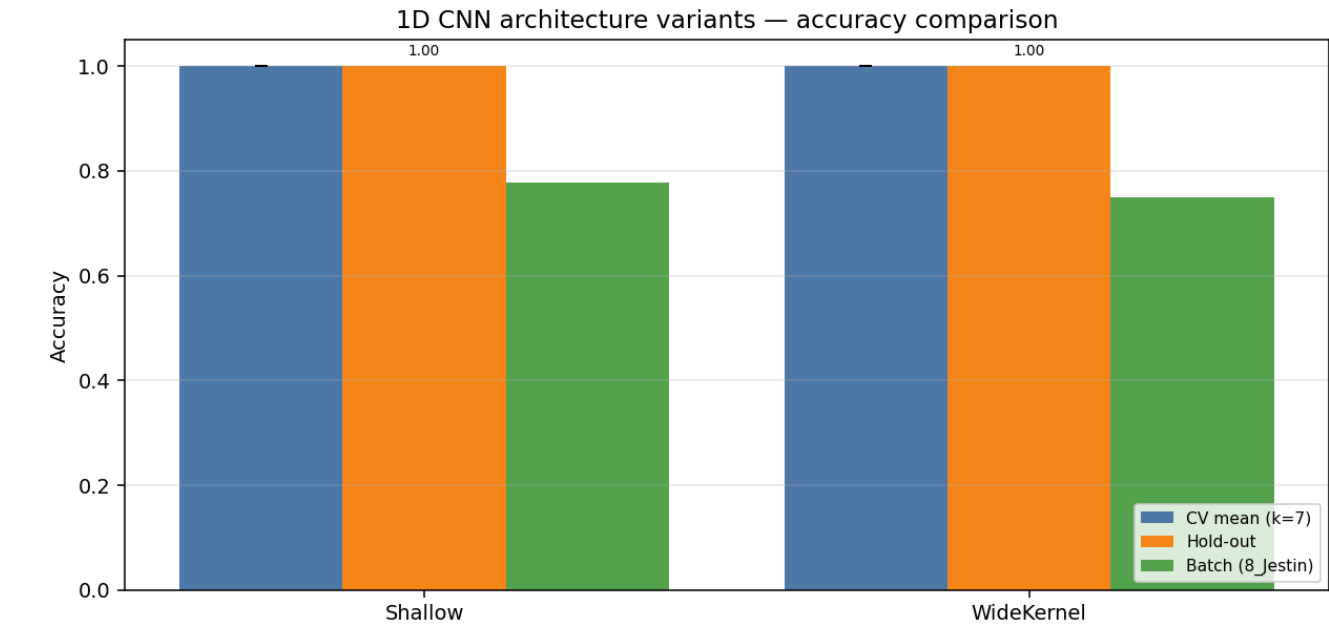
Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✓	—	—	—
left thumb	✓	✓	✓	✓
left index	✓	✓	✓	✓
left middle	✓	✓	✓	✓
left ring	✓	✓	✓	✓
left pinky	✓	✓	✓	✓
left wrist	✓	—	—	—
right palm	✓	—	—	—
right thumb	✓	✓	✓	✓
right index	✓	✓	✓	✓
right middle	✓	✓	✓	✓
right ring	✓	✓	✓	✓
right pinky	✓	✓	✓	✓
right wrist	✓	—	—	—

Enabled: 44 · Disabled: 0 · Modalities: YPR, Accel → 144 channels total.

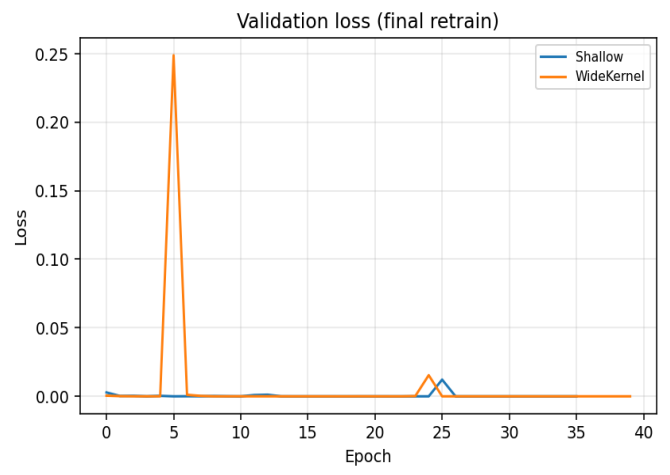
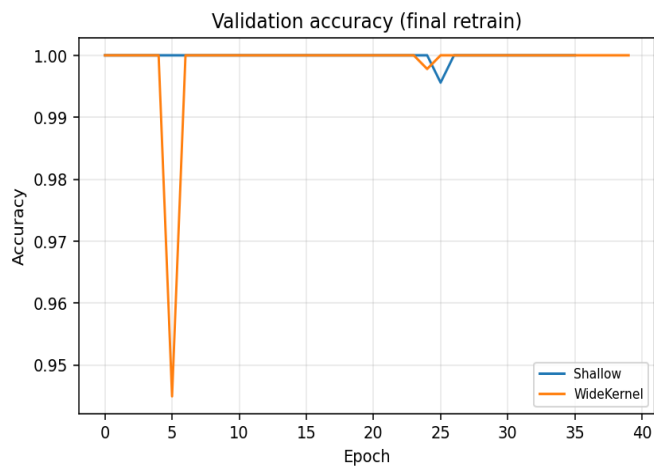
Variant comparison (summary)

Variant	Params	CV mean (k=7)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Shallow	67,334	1.0000	0.0000	1.0000	0.0000	0.7778	28/36
WideKernel	123,430	1.0000	0.0000	1.0000	0.0000	0.7500	27/36

Best on hold-out: **Shallow** · Best on CV: **Shallow**



Training curves (final retrain on full training pool)



Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	67,334
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.7778 (28/36)

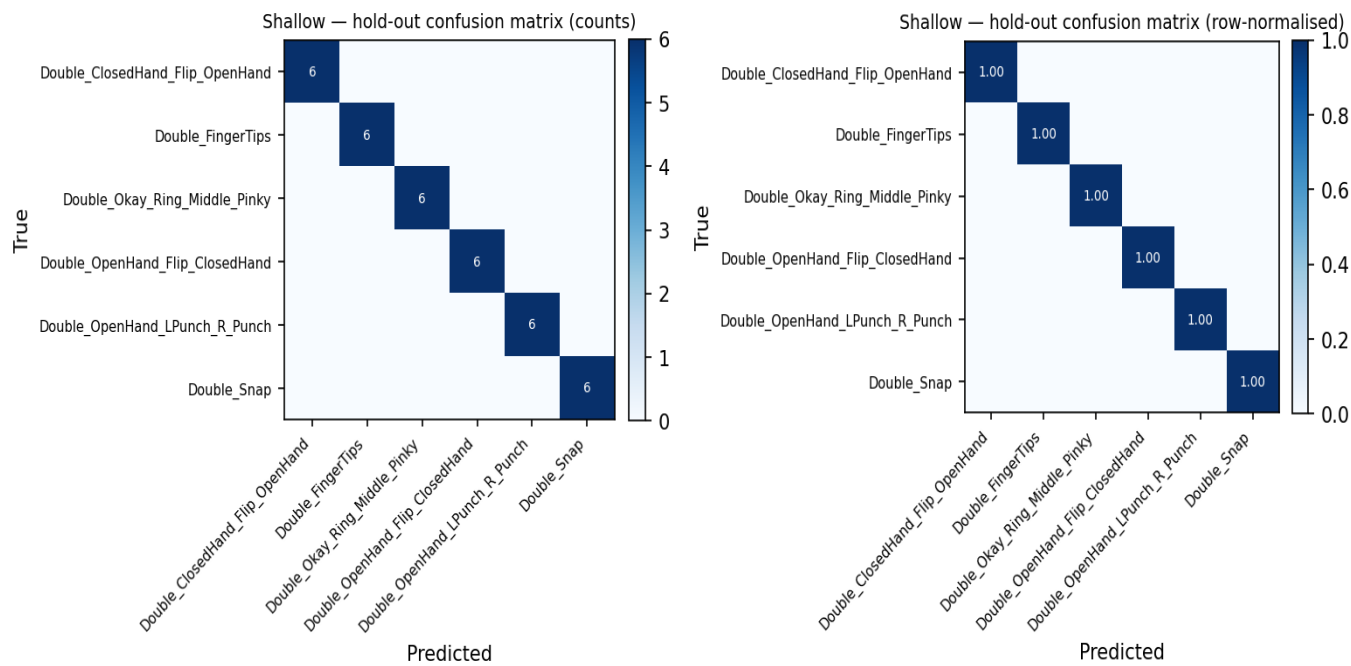
Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	0.9783
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	0.9783

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	1.000	1.000	1.000	36
weighted avg	1.000	1.000	1.000	36

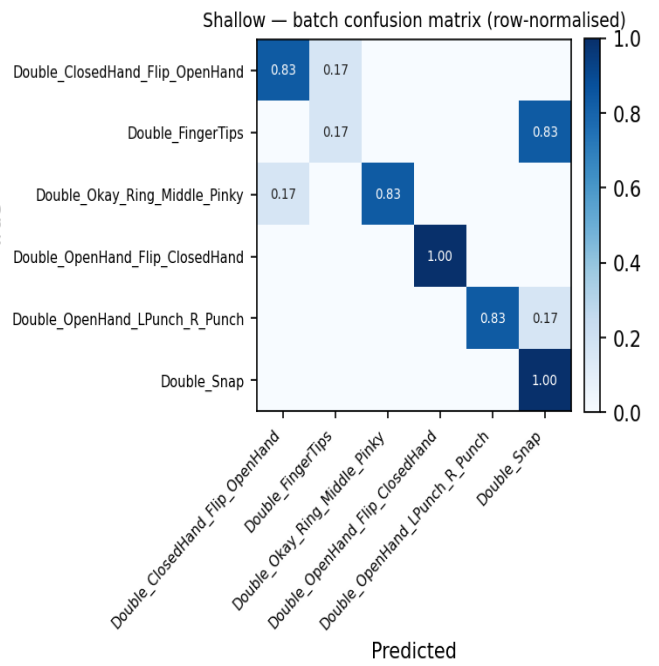
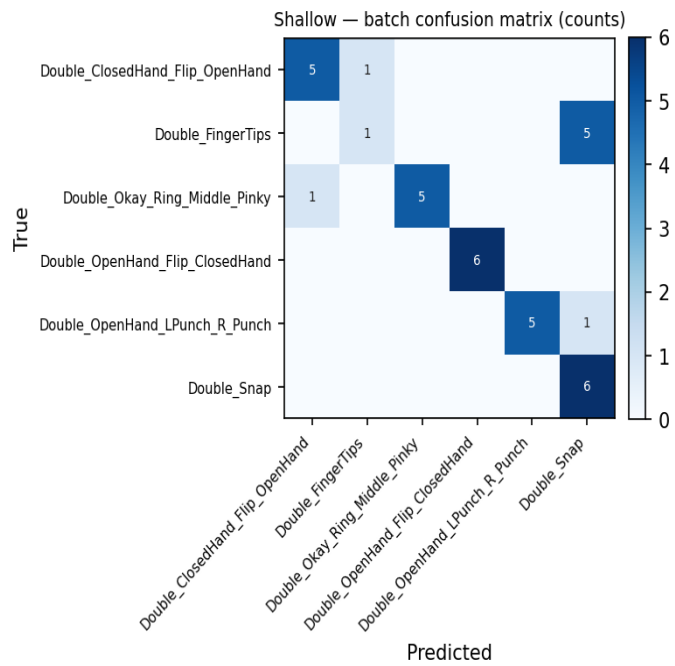
Hold-out confusion matrix



Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	1	6	0.167
Double_Okay_Ring_Middle_Pinky	5	6	0.833
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	5	6	0.833
Double_Snap	6	6	1.000
Overall	28	36	0.778

Batch confusion matrix (8_Jestin)



Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	123,430
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.7500 (27/36)

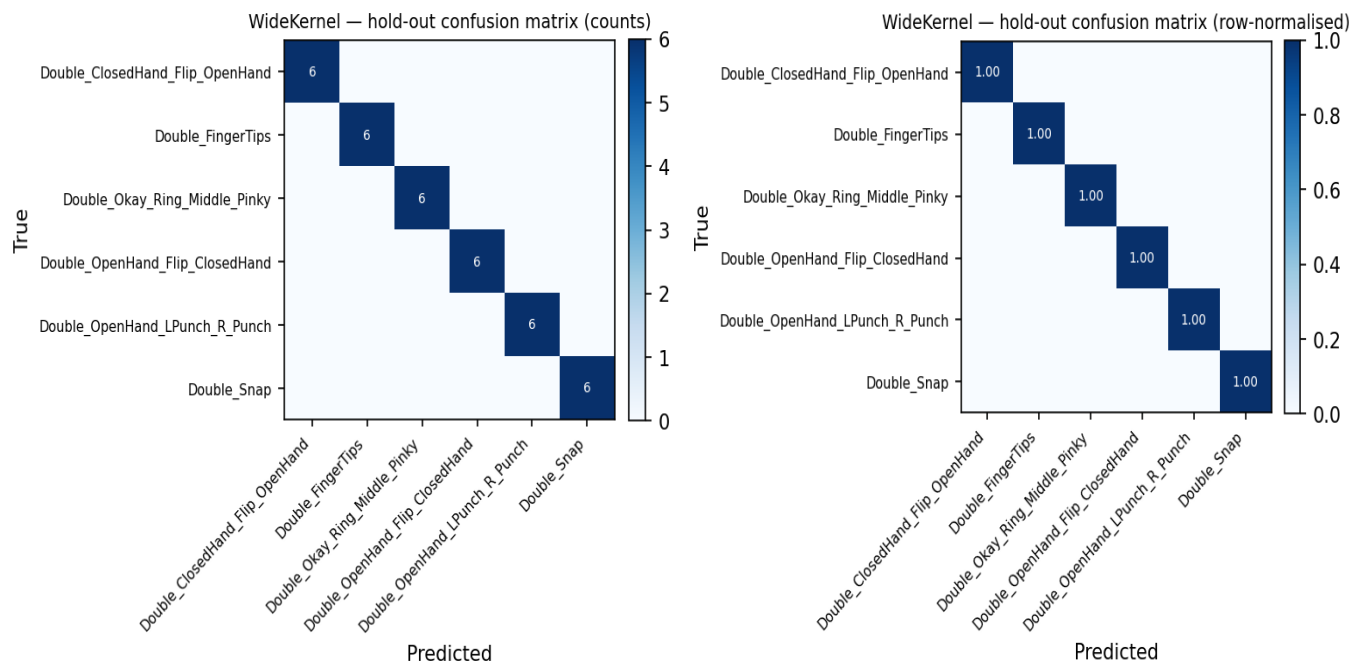
Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	1.000	1.000	1.000	36
weighted avg	1.000	1.000	1.000	36

Hold-out confusion matrix



Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	0	6	0.000
Double_Okay_Ring_Middle_Pinky	6	6	1.000
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	4	6	0.667
Double_Snap	6	6	1.000
Overall	27	36	0.750

Batch confusion matrix (8_Jestin)

